Retrieval Head Atlas: A Role-Decomposed Circuit for Semantic Long-Context Retrieval

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ABSTRACT

Long-context language models can often answer questions about information that appeared thousands of tokens earlier, but task accuracy alone does not explain how this retrieval happens internally. In our previous work, we built a retrieval-head atlas for Qwen2.5-1.5B-Instruct and showed that long-context copying behavior is supported by a sparse set of attention heads: only a small subset of heads reliably pointed to the source answer span, and targeted ablations showed that these heads played a causal role in difficult long-distance retrieval.

In this work, we extend that analysis from literal copying to semantic key-value retrieval. Instead of asking only whether a model can copy a hidden string, we test whether the same internal machinery supports retrieval when the query uses aliases, paraphrases, relational descriptions, and distractor-heavy contexts. We combine semantic retrieval probes with causal ablation, clean-to-corrupt activation patching, single-head decomposition, attention tracing, and activation-difference analysis across 8k and 16k token contexts.

The main finding is that semantic retrieval in Qwen2.5-1.5B-Instruct is not implemented by one undifferentiated set of important heads. It is better described as a role-decomposed circuit. A dominant answer-content head, L22H7, directly attends to the answer-bearing context, changes strongly when the answer identity changes, and restores the largest single-head portion of the correct-answer log probability when patched from a clean run into a corrupted run. A smaller companion head, L22H10, contributes additional answer signal. In contrast, a separate set of non-address support heads is strongly necessary under ablation but does not itself transplant answer identity under clean activation patching. This separation between content transport and support-state machinery remains stable across early, middle, and late needle positions and across 8k-16k contexts.

These results sharpen the original Retrieval Head Atlas from a head-discovery project into a mechanistic account of semantic long-context retrieval. The model appears to route answer identity through a small address/content pathway while relying on broader support heads to make that retrieval usable.

KEYWORDS

mechanistic interpretability, long-context retrieval, attention heads, activation patching, ablation, semantic retrieval

1 INTRODUCTION

Large language models are increasingly used on long inputs: documents, transcripts, logs, codebases, legal records, and multi-turn conversations. In these settings, a model must often answer a question using a specific fact that appeared far earlier in the context. From the outside, this can look simple: the model either retrieves the right answer or it does not. But from an interpretability perspective, that behavioral view is incomplete. A model can achieve high retrieval accuracy while hiding very different internal mechanisms.

The central question of this project is:

> When a transformer retrieves an answer from long context, which internal components actually carry and use that information?

Our previous submission began answering this question by building a Retrieval Head Atlas. We used synthetic long-context probes where a secret value was hidden inside an 8k-token prompt and the model was asked to reproduce it. The model achieved near-perfect accuracy, but the internal attention patterns were not uniform. A sparse cluster of heads, mostly in later layers, placed attention on the answer span at the moment the model generated the answer. Targeted ablations showed that disabling selected heads could selectively damage far-context retrieval, while matched random-head controls did not reproduce the same drop. This established a useful first result: long-context retrieval is behaviorally robust, but internally sparse and causally localized.

However, the previous work left a deeper question open. Literal copying is only one form of retrieval. Real retrieval often requires semantic matching: the query may refer to an entity through an alias, a paraphrase, a relation, or a description rather than an exact repeated string. A head that helps copy a visible token span may or may not be the same head that supports semantic retrieval. Similarly, an attention head can be important in several different ways: it may point to the source answer, carry the answer content, prepare the query state, stabilize downstream computation, or participate in a support role that is necessary but not answer-specific.

This report therefore shifts from asking "which heads matter?" to asking "what roles do the heads play?" The difference is important. A list of important heads is useful, but a circuit explanation should say how those heads divide labor. In this work, we test whether Qwen2.5-1.5B-Instruct uses a stable, role-decomposed circuit for semantic long-context retrieval.

Our main result is that it does. Across semantic variants, context positions, and context lengths, one head, L22H7, consistently behaves like the dominant answer-content carrier. It attends strongly to the answer-bearing span, its activation changes substantially when the answer changes, and patching this head from a clean prompt into a corrupted prompt restores much of the correct-answer probability. L22H10 acts as a smaller companion. Other support heads are also crucial, but in a different way: ablating them strongly damages performance, while patching them does almost nothing to restore the specific answer identity. This means the retrieval mechanism is not simply "all important heads carry the answer." Instead, answer transport and retrieval support are separable roles.

2 WHAT THE PREVIOUS SUBMISSION ESTABLISHED

The previous submission introduced the core methodology and the first version of the Retrieval Head Atlas. Its goal was to move beyond accuracy and inspect the model's internal retrieval behavior during a controlled long-context task.

The task was a deterministic needle-in-haystack setup. A six-digit secret was inserted into a long prompt, and the model was asked to output that secret. Two settings were emphasized: a far setting where the secret appeared early in the prompt, and a near setting where it appeared close to the query. The model solved both settings with near-perfect accuracy, which created a ceiling effect: ordinary accuracy could no longer reveal which internal components mattered.

To look inside the model, the previous work defined a retrieval event. At the generation step where the model produced the first answer token, each attention head was checked for whether it placed substantial attention mass on the answer span. A head was treated as retrieval-like when it pointed to the answer span on examples where the model answered correctly.

This mapping produced three important observations.

First, retrieval-like attention was sparse. Most heads did not behave like direct pointers to the answer span. The strongest retrieval-like activity was concentrated in a limited set of heads, especially in later layers.

Second, retrieval behavior was distance-sensitive. Some heads were active in near retrieval but not in far retrieval, while the difficult far setting emphasized a different subset of heads. This suggested that long-distance retrieval was not just the same computation as local recency-based access.

Third, causal testing mattered. The previous work did not stop at attention maps. It used targeted ablation to disable selected heads and compared them against layer-matched random-head controls. In the difficult far setting, ablating the selected heads reduced retrieval performance while the matched random controls did not show the same effect. This supported the claim that the mapped heads were not merely correlated with retrieval; they contributed causally.

The previous submission also explored activation patching and external LongBench-style validation. Those results were useful but less conclusive than the ablation result. In particular, patching showed that many internal sites could improve answer probability, which suggested broad representational redundancy. This became a motivation for the present work: instead of treating all important heads as one group, we needed to separate different causal roles more carefully.

The previous report's headline result can be summarized as follows. On the 8k-token synthetic retrieval probe, the model solved both far and near retrieval almost perfectly, so ordinary accuracy made the task look solved. The internal analysis told a different story. The strongest retrieval-like heads were sparse and concentrated in late-middle to late layers. When the selected top-k heads were ablated, far retrieval dropped from perfect performance to roughly 94%, while a layer-matched random control stayed at perfect performance. Near retrieval remained robust. This gave the previous submission its central message: long-context retrieval can look effortless at the output level while depending on a small, distance-sensitive internal head ensemble.

The previous patching result was also important, but in a different way. Patching selected heads from a clean run into a corrupted run increased the correct-answer log probability, showing that internal activations could repair the output distribution. However, random layer-matched patch sets sometimes repaired even more strongly than the originally selected top-k set. Rather than treating this as a contradiction, the current report treats it as a clue: necessity and sufficiency may live in different parts of the circuit, and a head can be useful for repair without being the exact component whose ablation most damages retrieval. This is one of the reasons the current work focuses on role decomposition.

In short, the previous submission established the foundation:

- Long-context retrieval can be behaviorally perfect while internally sparse.
- A small group of heads shows retrieval-like attention to the answer span.
- Targeted ablations provide causal evidence that selected heads matter for far-context retrieval.
- Patching results suggest that necessity and sufficiency are not the same thing, so a deeper role-based decomposition is needed.

3 CURRENT STUDY: FROM RETRIEVAL HEADS TO RETRIEVAL CIRCUITS

The current study extends the previous work in three ways.

First, it moves from literal copying to semantic retrieval. The new prompts still contain a recoverable answer in a long context, but the query does not always match the context literally. We test five variants: literal, alias, paraphrase, relational, and distractor-heavy retrieval. This lets us ask whether the retrieval heads are merely copying exact strings or whether they support broader semantic key-value retrieval.

Second, it decomposes retrieval into functional roles. We use attention tracing to identify heads that directly attend to the answer span, ablation to test which groups are necessary, clean-to-corrupt activation patching to test which groups carry transplantable answer information, and activation-difference analysis to test which heads actually change when the answer identity changes. These tools deliberately answer different questions. A head can be necessary without being sufficient; it can attend to the answer without carrying a cleanly patchable answer representation; it can change activation without being causally important. The report treats these distinctions as central rather than incidental.

Third, it tests whether the mechanism generalizes across position and length. The core suite is run at 8k and 16k token contexts, with the answer placed early, in the middle, and late in the context. This prevents the main claim from depending on one convenient prompt position or one context length.

The current study is organized around four research questions:

1. Do heads discovered from literal long-context retrieval remain important for semantic retrieval?
2. Can the semantic retrieval mechanism be decomposed into address/content heads and support heads?
3. Is there a single dominant head that carries most of the transplantable answer identity?
4. Does the resulting circuit remain stable across 8k-16k contexts and across different answer positions?

The short answer to all four is yes, with useful nuance. The semantic retrieval circuit is broader than the initial literal-copying shortlist, but it has a stable structure. L22H7 is the dominant answer-content head. L22H10 is a smaller companion. Non-address support heads, especially query-tail and broader core heads, are strongly necessary but weak as clean answer donors. This gives the final report a stronger narrative than the previous submission: we are no longer only mapping retrieval heads; we are explaining how different heads divide labor inside a semantic retrieval circuit.

4 CONTRIBUTIONS

This work makes the following contributions.

1. It extends the Retrieval Head Atlas from literal copying to semantic key-value retrieval. The experiments show that the relevant heads are not limited to exact string matching; they remain important across alias, paraphrase, relational, and distractor-heavy prompts.
2. It identifies a stable role decomposition inside the retrieval circuit. Direct answer-address heads and non-address support heads behave differently under attention tracing, ablation, patching, and activation-difference analysis.
3. It isolates L22H7 as the dominant answer-content head. Across all tested settings, L22H7 has strong answer/needle attention, large clean-corrupt activation differences, and the largest single-head patch effect.
4. It shows that necessity and sufficiency separate cleanly. Non-address support heads are strongly necessary under ablation but do not meaningfully transplant answer identity when patched, while address/content heads are partially sufficient.
5. It demonstrates stability across context length and answer position. The main role-decomposition result holds across 8k and 16k contexts and across early, middle, and late answer placements.
6. It provides a reusable experimental pipeline and report asset set: semantic probes, functional group ablations, component patching, single-head patching, attention tracing, activation-difference analysis, summary tables, and report-ready figures.

5 REPORT ORGANIZATION

The remainder of the report follows the same broad conference-style structure used in the previous submission: background, experimental setup, methods, results, discussion, limitations, and conclusion. The scientific story is different, but the reader experience stays close to the university/jury format.

Section 6 summarizes background and related work: attention heads, retrieval heads, long-context retrieval, causal ablation, and activation patching.

Section 7 describes the experimental setup: model, semantic prompt variants, clean/corrupt prompt pairs, context lengths, answer positions, and metrics.

Section 8 describes the methods: attention tracing, group ablation, clean-to-corrupt patching, single-head decomposition, activation-difference analysis, and matched controls.

Sections 9-12 present the results: semantic retrieval heads remain causal, the circuit separates into address/content and support roles, L22H7 dominates answer-content transport, and the pattern generalizes to 16k contexts.

Sections 13-15 discuss interpretation, limitations, future work, and the final conclusion.

6 BACKGROUND AND RELATED WORK

6.1 Circuit inspiration and related work

This project is inspired by the mechanistic interpretability view that neural networks can sometimes be understood as circuits: interacting internal components that implement a recognizable computation. In this framing, a model is not only a black-box function from prompt to answer. It is also a collection of internal pathways, some of which can be localized, intervened on, and tested.

The transformer-circuits line of work is especially relevant here. That work studies how attention heads, MLPs, residual streams, and token-level features interact to implement behaviors such as induction, copying, and in-context pattern completion. The key methodological lesson is that individual heads are rarely meaningful in isolation. A head becomes interpretable when we understand what it reads, what it writes, and how its output is used by later components.

Induction-head work is a useful analogy. Induction heads are not merely heads with visually interesting attention patterns; they are components that implement a specific algorithmic behavior: detecting a repeated prefix and predicting the next token from an earlier occurrence. The circuit idea is that a capability can be decomposed into roles, and those roles can be tested mechanistically. Our retrieval-head project starts from a similar intuition, but the behavior is long-context answer retrieval rather than short-pattern induction.

Another relevant line of work studies causal tracing and activation patching [3]. These methods compare clean and corrupted runs of a model and patch internal activations to identify where information relevant to the output is stored or transported. This is the source of our clean-to-corrupt patching setup: if a head carries answer identity, then copying that head's clean activation into a corrupt prompt should increase the clean-answer probability.

Recent retrieval-head work [8] directly motivates the previous submission and this continuation. It argues that sparse attention heads can play a mechanistic role in long-context factuality by retrieving relevant earlier facts. Our previous report adapted that idea into a controlled Retrieval Head Atlas for Qwen2.5-1.5B-Instruct. The present report extends the idea further: it asks whether retrieval heads form a role-decomposed semantic retrieval circuit, rather than merely a shortlist of heads that point to answer spans.

Long-context evaluation work is also part of the motivation. Studies such as “lost in the middle” show that models’ use of long context can depend strongly on where information appears. This matters for interpretability because a mechanism discovered at one answer position may not generalize. That is why the current experiments test early, middle, and late answer positions and extend from 8k to 16k contexts.

This report therefore sits at the intersection of several related lines of work: Transformers [6], transformer circuits [1], induction heads [4], causal tracing and activation patching [3], retrieval heads [8], Qwen2.5 [5], and long-context position-sensitivity studies such as Lost in the Middle [2].

6.2 Attention heads and long-context retrieval

Transformers process a sequence by repeatedly mixing information across token positions. The attention mechanism is the part of the model that decides, at each layer and token position, which earlier tokens should influence the current representation. In a multi-head attention layer, each attention head computes its own attention pattern and value mixture. This means that a single transformer layer does not have one monolithic way of looking backward through context; it has several parallel attention heads that can specialize in different patterns.

This structure makes attention heads a natural starting point for studying long-context retrieval. If a model answers a question using information that appeared thousands of tokens earlier, one possible internal mechanism is that some attention heads directly attend back to the relevant source span when the answer is being produced. Such heads can be interpreted as pointer-like components: they do not merely store information locally, but connect the current query position to a distant part of the context.

However, attention is not the whole computation. An attention head has at least two distinct aspects: where it attends and what information it writes back into the residual stream. A head can look at the right source position but write information that is not decisive for the output. Conversely, a head may be causally important without directly attending to the answer span, because it prepares the query representation, stabilizes the residual stream, or supports downstream heads. This distinction is central to the present report. We do not assume that “attention to the answer” and “causal responsibility for the answer” are the same thing.

6.3 Retrieval heads

Recent interpretability work has argued that long-context factual retrieval can depend on a sparse set of retrieval heads: attention heads that attend to relevant earlier facts and whose intervention affects factual output. This idea is attractive because it gives a concrete mechanism for a capability that otherwise looks diffuse. Instead of saying only that “the model uses the context,” we can ask which heads point to the relevant context and whether those heads matter causally.

Our previous submission followed this retrieval-head framing. It showed that, on a controlled needle-in-haystack task, Qwen2.5-1.5B-Instruct solved long-context copying with near-perfect accuracy while only a small subset of heads showed direct retrieval-like attention. This supported the idea that retrieval behavior can be internally sparse even when the model’s external behavior looks robust.

The present work extends that framing in two directions. First, it studies semantic retrieval rather than only literal copying. In realistic retrieval settings, the query often refers to information indirectly: through aliases, paraphrases, descriptions, relations, or nearby distractors. A mechanism that only copies exact strings would be less interesting than one that supports semantic key-value retrieval. Second, it moves from identifying retrieval heads to decomposing a retrieval circuit. The question is not merely whether some heads matter, but what different heads contribute.

6.4 Mechanistic interpretability: from observation to intervention

A common risk in interpretability is over-reading observational evidence. Attention maps can be visually compelling, but an attention map alone does not prove that a head is necessary for the model’s answer. A head may attend to the answer span because the answer is already represented elsewhere, or because attention to that span is correlated with success but not required for it.

For this reason, the project uses interventions. Interventions change internal model components and measure the effect on output probabilities. In this report, the main output quantity is the log probability of the correct answer under a controlled prompt. If an intervention lowers the correct-answer log probability, that suggests the intervened component contributed to the behavior. If an intervention restores the correct-answer log probability in a corrupted setting, that suggests the patched component carries useful information for the behavior.

The important point is that different interventions answer different questions. We use ablation, patching, attention tracing, and activation-difference analysis together because no single tool is sufficient on its own.

6.5 Ablation as a necessity test

Ablation disables a selected internal component and measures how much the model’s behavior changes. In the context of attention heads, ablation usually means removing or zeroing a head’s contribution at a particular position or stage of the forward pass. If ablating a head or group of heads reduces the correct-answer log probability, this is evidence that the component was necessary for the original computation.

Necessity should be interpreted carefully. A small ablation effect does not always mean a component is irrelevant. Transformer circuits can be redundant: several heads may perform overlapping roles, so removing one head may have little effect while removing a coordinated group has a large effect. Conversely, a large ablation effect does not necessarily mean the ablated component directly carried the answer. It may have supported the computation in another way.

This is why the report uses both single-head and group-level ablations. Single-head sweeps help identify candidate heads and localize effects. Functional group ablations test whether a role-defined group is necessary as a system. Matched inactive controls are included to reduce the risk that a result is caused merely by ablating many heads in late layers.

6.6 Activation patching as a sufficiency test

Activation patching asks a different question. Instead of removing a component, we copy an activation from one run of the model into another run. In this project, the most important patching setup is clean-to-corrupt patching. The clean prompt contains one answer; the corrupt prompt is identical except that the answer value is changed. The corrupt prompt makes the model prefer the corrupt answer. We then copy selected clean activations into the corrupt run and measure whether the model becomes more likely to output the clean answer.

If patching a head increases the clean-answer log probability, that head contains information that is at least partially sufficient to restore the answer in that patching setup. This is a stronger kind of evidence than attention alone because it tests whether the internal state can causally move the output distribution.

At the same time, patching has its own limitations. A successful patch does not prove that a component is uniquely responsible for the behavior; many sites may carry overlapping answer information. A failed patch also does not prove that a component is unimportant; the component may be necessary for setting up the computation while not carrying a cleanly transplantable answer representation at the patched site. For this reason, patching and ablation must be read together.

6.7 Why necessity and sufficiency can disagree

One of the main conceptual lessons of this work is that "necessary" and "sufficient" are not the same property. A support component can be necessary because the model needs it to maintain a useful query state, route residual information, or stabilize the computation. But that same component may not carry the identity of the answer. If we patch it from a clean run into a corrupt run, it may not restore the clean answer because there is little answer-specific information to transplant.

The reverse can also happen. A component can be a good patch target because it contains answer-specific information, while being less damaging under ablation because other components can partly compensate when it is removed. In a redundant neural system, ablation and patching are complementary views rather than interchangeable tests.

This distinction is the key reason the present report moves beyond a simple "important heads" list. The results show that some heads behave like answer-content heads: they attend to the answer span, change strongly when the answer changes, and restore answer probability when patched. Other heads behave like support heads: they are strongly necessary under ablation but weak as answer donors under patching. The circuit explanation depends on separating these roles.

6.8 Semantic retrieval and position generalization

Long-context retrieval is not a single fixed problem. Retrieval can be easy when the answer appears near the query and harder when it appears far away. It can also change when the query refers to the answer literally versus semantically. A robust mechanistic claim should therefore not depend on one prompt shape or one answer position.

This report evaluates semantic retrieval across five prompt variants: literal, alias, paraphrase, relational, and distractor-heavy. It also tests early, middle, and late answer positions at both 8k and 16k token context lengths. This matters because a head that appears important only at one position or one context length may reflect a local artifact rather than a stable retrieval mechanism.

The goal is not to claim that the discovered circuit is universal across all models or all retrieval tasks. The goal is narrower and more defensible: to show that, within Qwen2.5-1.5B-Instruct and this controlled semantic retrieval family, the same role-decomposed mechanism appears repeatedly across meaningful variations in wording, position, and length.

7 EXPERIMENTAL SETUP

7.1 Model

All experiments in the current study use Qwen2.5-1.5B-Instruct [5], an instruction-tuned transformer language model with 28 layers and 12 attention heads per layer. This gives 336 attention heads in total. The model is small enough to run repeated mechanistic experiments on a single GPU instance, but large enough to show non-trivial long-context retrieval behavior.

The experiments are implemented in PyTorch using Hugging Face Transformers [7]. The model is loaded through the repository configuration as `Qwen/Qwen2.5-1.5B-Instruct`, with automatic precision selection unless otherwise specified. The main GPU execution path is designed for the SageMaker notebook environment used for the project. Local execution on the Mac is used mainly for code editing, artifact analysis, table generation, and report writing.

7.2 Semantic retrieval prompt family

The current experiments use a controlled semantic retrieval task. Each prompt contains a long synthetic document, a marked answer-bearing span, irrelevant filler text, and a final question. The answer is always a six-digit value, and the model is instructed to answer with only the six digits.

The prompt structure is intentionally controlled. The answer span is surrounded by explicit markers, `[NEEDLE_START]` and `[NEEDLE_END]`, which allow the analysis scripts to locate the source span for attention tracing. The surrounding context consists of deterministic filler words. This synthetic setup does not aim to imitate a natural document perfectly; its purpose is to create a clean environment where the answer location, answer value, distractor values, and prompt length are known exactly.

The dataset uses five retrieval variants:

1. **Literal.** The document states the access code directly, and the question asks for the access code.
2. **Alias.** The document assigns a key to a codename, and the question asks for the key associated with that codename.
3. **Paraphrase.** The document describes the answer as a six-digit sequence needed to unlock an archive, and the question asks for that sequence.
4. **Relational.** The document links a person to an object and then gives that person's badge number; the question asks for the badge number of the person who owns the object.

5. **Distractor-heavy.** The document contains a decoy ticket number and a true access code, and the question asks for the true code associated with a codename.

These variants are designed to test more than exact string copying. In the alias, paraphrase, and relational cases, the model must connect the final query to the relevant fact through meaning rather than through a repeated phrase alone. In the distractor-heavy case, the model must avoid a nearby irrelevant six-digit number.

7.3 Prompt length and answer position

Prompts are calibrated to target two context lengths:

- 8k tokens, represented by `target_tokens = 8192`
- 16k tokens, represented by `target_tokens = 16384`

For each target length, the amount of filler text is adjusted so that the final tokenized prompt length is very close to the target. In the committed 16k runs, mean prompt length was approximately 16,384 tokens, with observed minimum and maximum lengths within about one token of the target. This matters because the length-generalization claim should be based on actual long prompts, not on a nominal setting that silently produces shorter inputs.

The answer span is placed at three position fractions:

- `needle_frac = 0.1`: early in the document, far from the final query
- `needle_frac = 0.5`: near the middle of the document
- `needle_frac = 0.9`: late in the document, closer to the final query

This gives six main context settings: 8k/0.1, 8k/0.5, 8k/0.9, 16k/0.1, 16k/0.5, and 16k/0.9. Each setting uses 40 examples: five semantic variants times eight examples per variant.

7.4 Clean and corrupt prompt pairs

Activation patching requires paired prompts. For each clean prompt, a corrupt prompt is created by replacing the six-digit answer with a different six-digit value while keeping the rest of the prompt structure fixed. The corrupt answer is selected so that the clean and corrupt prompts have matching token lengths under the tokenizer. This prevents trivial alignment problems during activation patching.

The clean prompt asks for the clean answer, and the corrupt prompt asks for the corrupt answer in the same semantic format. The patching experiment then measures whether copying selected activations from the clean run into the corrupt run increases the model's log probability for the clean answer. This setup creates a controlled causal question: does the patched component carry information about the clean answer identity?

The patching baseline records three quantities:

- the clean prompt's log probability for the clean answer,
- the corrupt prompt's log probability for the clean answer,
- the corrupt prompt's log probability for the corrupt answer.

The second and third values verify that the corrupt prompt really changes the model's expected answer. In the main 8k early-position run, for example, the clean answer is much less likely under the corrupt prompt, while the corrupt answer is likely under the corrupt prompt. This creates a meaningful recovery gap for patching.

7.5 Head groups

The experiments use several head groupings.

The first grouping comes from the previous retrieval-head atlas and early semantic ablation probes. These heads are used to test whether literal-retrieval heads remain relevant under semantic variants.

The second grouping comes from a broader semantic neighborhood sweep. This sweep includes patch-ranked heads, earlier candidate heads, and heads sampled by random controls that turned out to be more damaging than expected. The purpose of the sweep is to identify a broader semantic retrieval core rather than rely only on the first top-k list.

The final grouping is functional. Heads are grouped according to their observed role:

- **Answer-address heads.** Heads that directly attend to the answer or needle span. The key heads in this group are L22H7, L22H10, and L21H11.
- **Non-address core heads.** Heads that are necessary under ablation but do not directly attend to the answer span.
- **Query-tail support heads.** Heads whose attention behavior concentrates near the query tail and whose ablation strongly affects correct-answer probability.
- **First-token/sink heads.** Heads with sink-like attention patterns that are nevertheless functionally important.
- **Inactive controls.** Layer-matched or role-matched heads selected to test whether effects are specific to the active groups.

These groups are not assumed in advance. They are constructed from the sequence of experiments: semantic ablation, neighborhood sweep, attention tracing, and functional controls.

7.6 Main metrics

The report uses log probabilities rather than only exact-match accuracy. This is important because the model often answers correctly in the baseline condition, making accuracy insensitive to smaller causal effects. Log probability provides a smoother measure of how strongly the model prefers the correct answer.

The main metrics are:

- **Baseline log probability.** The mean log probability assigned to the correct answer before intervention.
- **Ablation delta.** The change in correct-answer log probability after disabling a selected head or group. Negative values mean the intervention hurt the correct answer.
- **Patch delta.** The change in clean-answer log probability after patching clean activations into a corrupt prompt. Positive values mean the patch restored some clean-answer probability.
- **Recovery fraction.** The patch delta divided by the available clean-vs-corrupt recovery gap. This normalizes patch strength by how much recovery was possible.
- **Gold attention mass.** The attention mass a selected head places directly on the gold answer tokens.

- **Needle attention mass.** The attention mass a selected head places on the full marked answer-bearing span.
- **Activation relative difference.** The clean-vs-corrupt activation difference normalized by the clean activation norm. This measures how strongly a component changes when the answer identity changes.

Together, these metrics let us distinguish four types of evidence: behavioral preference, causal necessity, causal sufficiency, and answer-specific representational change.

8 METHODS

8.1 Overview

The method follows the same broad philosophy as the previous Retrieval Head Atlas pipeline: first identify candidate heads, then test them with causal interventions. The difference is that the current work does not stop at one head ranking. It uses multiple complementary tests to separate different roles inside the retrieval mechanism.

The full pipeline is:

1. Run a semantic ablation probe on the initial retrieval-head candidates.
2. Expand to a broader semantic neighborhood when random controls reveal additional active heads.
3. Trace attention to identify which heads directly attend to the answer span.
4. Build functional groups: answer-address heads, non-address support heads, query-tail heads, sink-like heads, and inactive controls.
5. Run group ablations to test necessity.
6. Run clean-to-corrupt component patching to test sufficiency.
7. Run patch-then-ablate interaction controls to test whether support heads gate use of the address-head patch.
8. Run activation-difference analysis to measure which heads actually change with answer identity.
9. Run single-head patching to identify the main answer-content donor.
10. Repeat the core suite across context positions and context lengths.

This section describes each step in enough detail to make the later causal claims auditable.

8.2 Scoring answer probability

Most experiments measure the model's preference for the correct answer using teacher-forced mean answer log probability. Let a prompt be tokenized as x, and let the gold answer be tokenized as $y = (y_1, \ldots, y_m)$. The score is:

$$\ell(x, y) = \frac{1}{m} \sum_{t=1}^{m} \log p_\theta(y_t \mid x, y_{<t}).$$

This score is computed at the final query point and then through the answer tokens under teacher forcing. We use mean log probability rather than total log probability so that scores remain comparable when tokenization differs slightly across six-digit answers.

For interventions, the same score is recomputed with selected internal components modified. The key readout is a change in log probability:

$$\Delta\ell = \ell_{\text{intervened}}(x, y) - \ell_{\text{baseline}}(x, y).$$

For ablations, a negative $\Delta\ell$ means the intervention made the correct answer less likely. For patching, a positive $\Delta\ell$ means the patch restored some probability for the clean answer.

8.3 Query-step interventions

The strongest current experiments intervene at the query step. This means the prompt is first prefetched into the KV cache, and the intervention is applied when the model processes the final prompt token immediately before scoring the answer. This design asks which components matter at the moment the model must use the long context to answer the final question.

The intervention site is the input to the attention output projection (`o_proj`) for selected heads. In a transformer layer, the attention module produces per-head vectors that are concatenated and passed through the output projection. If the hidden size is d, the number of heads is H, and each head has dimension $d_h = d/H$, then the vector entering the output projection can be viewed as:

$$z^{(\ell)} = [z_1^{(\ell)}; z_2^{(\ell)}; \ldots; z_H^{(\ell)}].$$

Here $z_h^{(\ell)}$ is the slice corresponding to head h in layer ℓ. Ablation and patching operate on these slices.

This intervention target is useful because it directly modifies what an attention head writes into the residual stream after it has computed its attention pattern and value mixture. It is also a limitation: it does not separately isolate query, key, value, attention-logit, or value-vector mechanisms. That limitation is discussed later.

8.4 Head ablation

For a selected set of heads $S = \{(\ell, h)\}$, ablation zeros the corresponding head slices at the intervention site:

$$z_h^{(\ell)} \leftarrow 0 \quad \text{for all } (\ell, h) \in S.$$

The ablation effect is:

$$\Delta_{\text{ablate}}(S) = \ell_{\text{ablate}(S)}(x, y) - \ell_{\text{base}}(x, y).$$

A more negative value means that the selected heads were more necessary for the model's correct-answer probability. The report uses both single-head ablations and group ablations. Single-head ablations help identify candidate heads. Group ablations test whether a functional subsystem is necessary as a coordinated unit.

The main functional group ablations are run query-only. This keeps the interpretation focused: we are asking what happens when a head's contribution is removed at the final retrieval/use step, rather than throughout every answer-token continuation step.

8.5 Layer-matched and inactive controls

Ablating heads in late layers can have different effects from ablating heads in early layers. Therefore, random controls must be matched by layer. If the active group contains k_ℓ heads in layer ℓ, the layer-matched random control also samples k_ℓ heads from the same layer, excluding the active heads when possible. This prevents an unfair comparison where the active group appears important only because it occupies sensitive layers.

The project uses two kinds of controls:

1. **Layer-matched random controls.** These are used in the early semantic ablation probe to test whether the initial top-k heads are more damaging than random heads from the same layers.
2. **Inactive functional controls.** These are used in later group experiments. They are chosen to match the rough layer/role structure of active groups but are empirically weak or inactive in the relevant prior sweeps.

The inactive controls matter because some active groups are large. For example, the non-address core has more heads than the answer-address group. A large ablation effect must therefore be compared against controls that help rule out a simple "more heads removed" explanation.

8.6 Semantic ablation probe

The first extension beyond the previous submission tests whether retrieval heads discovered from literal long-context behavior remain important under semantic variants. The probe uses five variants: literal, alias, paraphrase, relational, and distractor-heavy. For each example, the model is scored under three conditions:

- baseline,
- ablation of the selected top-k heads,
- ablation of layer-matched random heads.

For each example, the script records:

$$\Delta_{\text{topk}} = \ell_{\text{topk ablated}} - \ell_{\text{base}},$$

and

$$\Delta_{\text{random}} = \ell_{\text{random ablated}} - \ell_{\text{base}}.$$

The early run showed that the top-k heads remained consistently damaging across semantic variants, while random controls were weaker on average. However, the random controls were not always inert. Some random draws were unexpectedly damaging. Rather than ignore this, we treated it as evidence that the initial top-k list was incomplete: the random draws were sometimes sampling real members of a broader semantic retrieval circuit.

This led to the neighborhood sweep.

8.7 Neighborhood single-head sweep

The neighborhood sweep broadens the candidate set beyond the original top-k heads. It includes:

- heads from the previous patch-ranked and retrieval-event lists,
- heads from the semantic single-head sweep,
- heads from earlier near/far mapping artifacts,
- heads that appeared in damaging random-control draws.

Each head is ablated individually at the query step and ranked by its mean effect on correct-answer log probability. The goal is not to claim that single-head ablation fully explains the circuit. Instead, the sweep is used to discover a broader semantic core and to separate active circuit members from inactive controls.

This step is important narratively. What initially looked like a random-control complication became a discovery mechanism. The semantic retrieval circuit was broader than the first retrieval-head shortlist, and several important support heads were found through this expanded search.

8.8 Attention tracing

Attention tracing asks where selected heads look at the query step. The prompt contains explicit marker strings around the answer-bearing span. The tokenizer offset mapping is used to convert character spans into token spans:

- G: the gold answer token span,
- N: the full needle span between `[NEEDLE_START]` and `[NEEDLE_END]`,
- D: distractor six-digit spans outside the gold answer.

For a head with query-step attention vector $a^{(\ell,h)}$, gold attention mass is:

$$M_G^{(\ell,h)} = \sum_{i \in G} a_i^{(\ell,h)}.$$

Needle attention mass is:

$$M_N^{(\ell,h)} = \sum_{i \in N} a_i^{(\ell,h)}.$$

The scripts also record distractor mass, gold rank, whether the attention argmax lies inside the gold span, and whether the argmax lies inside the needle span.

Attention tracing is observational, not causal. Its role is to distinguish direct address heads from heads that are necessary for other reasons. In the final interpretation, a direct answer-address head is one that both attends to the answer/needle span and has causal evidence from ablation or patching.

8.9 Functional group construction

After the neighborhood sweep and attention trace, heads are assigned to functional groups.

The most important groups are:

- **Answer-address.** Heads with direct attention to the answer-bearing span. This group contains L22H7, L22H10, and L21H11.
- **Non-address core.** Heads that are necessary under ablation but do not directly attend to the answer span.
- **Query-tail.** Heads whose attention behavior concentrates near the final query tokens and whose ablation strongly affects retrieval.
- **First-token/sink.** Heads with sink-like behavior that are nevertheless not inert under ablation.
- **Strong/core groups.** Larger aggregate groups used to measure total circuit necessity.
- **Inactive controls.** Heads selected as matched controls for active functional groups.

This grouping is not intended to be a perfect biological taxonomy of heads. It is an operational grouping based on converging experimental evidence. The report uses it to ask whether different roles show different causal signatures.

8.10 Component patching

Component patching tests whether selected heads carry answer-specific information that can be transplanted from a clean prompt to a corrupt prompt. Let x_c be the clean prompt with clean answer y_c, and let x_r be the corrupt prompt with corrupt answer y_r. The corrupt prompt is constructed so that the clean answer y_c becomes unlikely.

First, the clean query-step activations are cached:

$$z^{(\ell,h)}_{\text{clean}}.$$

Then, during the corrupt run, selected head slices are replaced:

$$z^{(\ell,h)}_{\text{corrupt}} \leftarrow z^{(\ell,h)}_{\text{clean}} \quad \text{for all } (\ell, h) \in S.$$

The patch effect is:

$$\Delta_{\text{patch}}(S) = \ell_{\text{patch}(S)}(x_r, y_c) - \ell_{\text{corrupt}}(x_r, y_c).$$

The available recovery gap is:

$$R = \ell_{\text{clean}}(x_c, y_c) - \ell_{\text{corrupt}}(x_r, y_c).$$

The recovery fraction is:

$$\rho(S) = \frac{\Delta_{\text{patch}}(S)}{R}.$$

If Δ_{patch} is positive, the patched component restored some probability for the clean answer. If ρ is large, the component restores a substantial fraction of the available clean-corrupt gap.

This is where the role decomposition becomes visible. Address heads have meaningful positive patch effects. Non-address support heads are strongly necessary under ablation but have very small patch effects, suggesting that they do not carry much answer identity at the patched site.

8.11 Patch-then-ablate interaction controls

After finding that answer-address patching helps while support-head patching is weak, the next question is whether support heads are required to use the restored address-head signal. To test this, the pipeline runs patch-then-ablate interactions.

For each example, it compares:

- corrupt baseline,
- address-head patch only,
- support-group ablation only,
- address-head patch plus support-group ablation.

The crucial quantity is the patch effect under ablation:

$$\Delta_{\text{patch under ablation}} = \ell_{\text{patch+ablate}}(x_r, y_c) - \ell_{\text{ablate only}}(x_r, y_c).$$

If this value collapsed relative to patch-only, it would suggest that the support group is required downstream to use the patched address-head information. The refined interaction controls showed a more nuanced result: support-group ablation strongly damages absolute clean-answer log probability, but the address-head patch still provides a positive boost after subtracting the ablate-only baseline. This argues against a strict serial dependency chain and supports a partly additive two-role mechanism.

8.12 Activation-difference analysis

Activation-difference analysis measures whether a component changes when the answer identity changes. For a selected group S, the query-step head slices are concatenated into one vector:

$$v_S = \text{concat}_{(\ell,h)\in S}\, z^{(\ell,h)}.$$

For each clean/corrupt prompt pair, we compute:

$$\Delta v_S = v_{S,\text{clean}} - v_{S,\text{corrupt}}.$$

The main reported metric is relative L2 difference:

$$r_S = \frac{\|v_{S,\text{clean}} - v_{S,\text{corrupt}}\|_2}{0.5(\|v_{S,\text{clean}}\|_2 + \|v_{S,\text{corrupt}}\|_2)}.$$

The scripts also record cosine similarity, mean absolute difference, and maximum absolute difference.

This analysis helps explain why necessity and sufficiency differ. Address heads, especially L22H7, have large clean-corrupt activation differences, indicating that they encode answer-specific state. Support heads are often nearly invariant between clean and corrupt prompts, even though ablation shows they are necessary. This supports the interpretation that support heads maintain retrieval/use state rather than carrying answer identity.

8.13 Single-head patch decomposition

The answer-address group contains three heads: L22H7, L22H10, and L21H11. Group patching shows that this group carries recoverable answer signal, but it does not reveal whether the signal is distributed evenly. Therefore, the pipeline runs single-head patching for:

- the three answer-address heads,
- several necessary support heads,
- an activation-sensitive but causally weak control,
- inactive controls.

The same patch delta and recovery fraction are computed for each single-head group. This experiment identifies L22H7 as the dominant answer-content donor: it accounts for most of the answer-address group patch effect across the tested settings. L22H10 is a smaller companion. L21H11 attends to the answer and is necessary, but is weak as a standalone clean donor.

8.14 Position and length generalization

Once the core role decomposition is established at 8k early-position prompts, the same suite is repeated across:

- early, middle, and late answer positions,
- 8k and 16k context lengths.

The repeated suite includes functional group ablation, functional component patching, single-head patching, attention tracing, and activation-difference analysis. This provides a stronger claim than a single setting would allow. The conclusion is not only that L22H7 matters in one probe, but that its attention, activation, and causal patch signatures remain stable across all tested position/length settings.

8.15 Statistical reporting

The report uses per-example rows as the basic unit of aggregation. For each metric and setting, the report tables include the mean, standard deviation, and an approximate 95% confidence interval:

$$\bar{x} \pm 1.96\frac{s}{\sqrt{n}}.$$

This normal-approximation interval is used as a readable sanity check rather than as a claim of exact distributional normality. For the main results, the direction and consistency of effects are more important than any single p-value. For example, L22H7 patching is positive with confidence intervals above zero in every tested length/position setting, and non-address core ablation is negative in 40/40 examples at every tested setting.

9 RESULTS I: SEMANTIC RETRIEVAL USES A CAUSAL HEAD NEIGHBORHOOD

The first question was whether heads discovered in the previous literal-retrieval setting remain important when retrieval becomes semantic. If the answer were no, then the previous atlas would be mainly a literal-copying result. If the answer were yes, then the same head neighborhood would be involved in a broader form of key-value retrieval.

9.1 Initial semantic ablation probe

The initial semantic probe used five variants: literal, alias, paraphrase, relational, and distractor-heavy. Each variant had eight examples, giving 40 examples total. The intervention ablated the selected top-k heads and compared them against layer-matched random heads. The most informative version used query-only intervention, because it focused the ablation on the final retrieval/use step.

The top-k ablation was consistently harmful. Across all 40 examples, the mean query-only top-k ablation effect was approximately:

$$\Delta_{\text{topk}} \approx -0.389.$$

A single layer-matched random draw was much weaker, near:

$$\Delta_{\text{random}} \approx -0.032.$$

This first result already showed that the selected heads were not merely literal-copy heads. They remained causally relevant when the query used aliases, paraphrases, relational descriptions, and distractor-heavy wording.

The variant-level pattern is important because it shows breadth rather than one lucky aggregate. In the query-only run, the top-k ablation hurt every variant:

The paraphrase case was especially strong, with a mean top-k delta around -0.518. This matters because paraphrase retrieval is less like exact copying than the literal condition. The top-k heads were therefore involved in more than just matching repeated surface text.

9.2 Stronger random controls revealed a broader circuit

The next control used 20 layer-matched random draws instead of a single random draw. This gave a more honest estimate of how often random heads from the same layers could damage retrieval.

The result was nuanced. The top-k group still hurt more than the average random draw:

$$\Delta_{\text{topk}} \approx -0.389, \qquad \mathbb{E}[\Delta_{\text{random draw}}] \approx -0.159.$$

The average gap was therefore:

$$\Delta_{\text{topk}} - \mathbb{E}[\Delta_{\text{random draw}}] \approx -0.231.$$

The top-k group hurt all 40 examples, and it was more damaging than each example's mean random draw in all 40 examples. This supports the original claim that the selected heads are meaningfully important.

However, the random-draw distribution was not fully inert. The most damaging random draw had a mean delta around -0.436, which was even more damaging than the top-k group. At first glance, this could look like a weakness in the result. But inspection showed that the damaging random draws often sampled heads from the same retrieval neighborhood, including heads that later appeared as active circuit members.

This changed the interpretation. The random controls were not simply noise; some of them were accidentally sampling real semantic-retrieval heads outside the initial top-k set. The correct conclusion was not that the top-k heads were unimportant, but that the initial top-k list was incomplete.

9.3 Single-head semantic sweep

To resolve this, we ran a query-only single-head ablation sweep over candidate heads. The sweep measured each head's individual effect on semantic retrieval and ranked heads by mean log-probability damage.

The first patch-ranked single-head sweep identified several individually harmful heads, including:

This sweep already showed that semantic retrieval was not driven by a single head. Several heads were consistently harmful when ablated, and their effects varied by semantic variant.

It also separated necessity from earlier patch rankings. Some heads that had appeared useful under patching were not strongly necessary as individual query-step ablations. This was an early warning that the final explanation would need separate notions of necessity and sufficiency.

9.4 Neighborhood sweep

The final selection run widened the sweep to 61 heads. This neighborhood included:

- the semantic single-head candidates,
- the patch-ranked candidates,
- heads from the previous near/far retrieval maps,
- non-zero ablation candidates from the earlier work,
- heads sampled by the 20 random draws.

This larger sweep produced 2,440 single-head ablation rows: 61 heads times 40 examples. The most harmful heads were:

Variant	Top-k ablation delta	Random-k ablation delta
Literal	-0.406	+0.025
Alias	-0.356	-0.075
Paraphrase	-0.518	-0.042
Relational	-0.283	-0.039
Distractor-heavy	-0.383	-0.027

Rank	Head	Mean ablation delta	Negative examples
1	L21H11	-0.110	35/40
2	L22H7	-0.107	37/40
3	L18H3	-0.087	40/40
4	L20H9	-0.055	35/40
5	L17H3	-0.043	32/40

Rank	Head	Mean ablation delta	Negative examples	Read
1	L20H7	-0.148	39/40	strongest newly discovered support head
2	L21H11	-0.110	35/40	answer-address candidate
3	L22H7	-0.107	37/40	answer-address/content candidate
4	L18H3	-0.087	40/40	broad support head
5	L17H10	-0.080	40/40	broad support head
6	L20H8	-0.075	37/40	support/query-related
7	L22H0	-0.073	40/40	support/query-related
8	L22H4	-0.069	40/40	support/query-related
9	L21H0	-0.068	40/40	support/sink-like
10	L20H6	-0.055	40/40	support head
11	L20H9	-0.055	35/40	semantic support
12	L21H3	-0.054	40/40	support/sink-like
13	L21H1	-0.052	40/40	support/sink-like

The strongest head in this broader sweep was L20H7, which was not the main answer-address head. This was a major clue. Semantic retrieval depended on a broader late-middle-layer circuit, not only on heads that directly pointed to the answer span.

The sweep also explained the earlier random-control complication. The effects of the 20 random draws correlated strongly with the sum of the single-head effects of the heads they sampled:

$$r \approx 0.92.$$

This means the damaging random draws were predictable from the active heads they accidentally included. The random controls were not mysterious; they revealed that the semantic retrieval circuit extended beyond the first top-k group.

9.5 Interpretation

The first result section establishes the starting point for the rest of the report:

1. Heads from the previous retrieval atlas remain causally relevant under semantic retrieval variants.
2. The semantic circuit is broader than the initial literal-copying top-k list.
3. Some random-control heads were damaging because they were not truly inactive; they were undiscovered circuit members.
4. The broader semantic core is concentrated mainly in layers 20-22, with support from neighboring layers such as 17, 18, and 23.
5. The strongest necessary heads are not all direct answer-address heads, motivating a functional decomposition.

This last point is the bridge to the next result. If all important heads simply attended to the answer span, the story would be straightforward: retrieval heads point to the answer and carry it forward. Instead, the most harmful single-head ablation in the neighborhood sweep was L20H7, which later attention tracing showed was not a direct answer-address head. Therefore, semantic retrieval requires at least two kinds of machinery: heads that address the answer and heads that support the retrieval computation without directly pointing to the answer.

10 RESULTS II: THE CIRCUIT SPLITS INTO ADDRESS HEADS AND SUPPORT HEADS

The neighborhood sweep showed that semantic retrieval depends on more than one head and more than one kind of head. The next question was whether these heads all perform the same job. Attention tracing and functional group ablation show that they do not. The circuit separates into heads that directly address the answer span and heads that are necessary for retrieval without directly pointing to the answer.

10.1 Direct answer-address heads

The attention trace identifies three heads with clear answer-address behavior: L22H7, L21H11, and L22H10. In the 8k early-position setting, their query-step attention masses are:

These values mean that the heads are not merely active in the same layer neighborhood; they are directly looking at the answer-bearing span at the moment the model must answer the query. L22H7 is especially clean: its attention argmax is inside the needle span in every example in this setting, and its gold attention mass is the highest among the traced heads.

This pattern remains visible across positions and lengths. At 16k, L22H7 needle attention is 0.865 at early position, 0.919 at middle position, and 0.931 at late position. This is important because it

Head	Gold attention	Needle attention	Argmax in needle
L22H7	0.665	0.848	1.000
L21H11	0.531	0.716	0.925
L22H10	0.355	0.702	0.700

rules out the possibility that L22H7 is only a short-context or one-position pointer.

10.2 Necessary heads that do not directly address the answer

Not all important heads behave this way. L20H7 is the clearest counterexample. It was the most harmful head in the broader single-head ablation sweep, with mean ablation delta around -0.148, but its attention trace shows almost no direct answer attention. In the 8k early-position setting, L20H7 has only 0.0015 gold attention mass and 0.0083 needle attention mass.

The same pattern appears for several other support heads. They can be strongly necessary under ablation while placing little direct attention on the answer tokens. Many of these heads attend near the query tail or to sink-like positions such as the first token. This suggests that semantic retrieval is not only an address operation. The model also needs support machinery that prepares, stabilizes, or routes the computation so that retrieved information can affect the output.

This is the first major role split:

- **Address heads** directly point to the answer-bearing context.
- **Support heads** are necessary for retrieval but do not themselves directly point to the answer.

10.3 Functional group ablation

To test whether these roles matter causally, heads were grouped into functional sets and ablated at the query step. The most important groups are:

- `answer_address`: L22H7, L22H10, L21H11
- `non_address_core`: necessary non-address heads from the semantic core
- `query_tail`: heads concentrated around query-tail behavior
- `first_token_sink`: sink-like heads that appeared functionally active
- inactive controls matched to active groups

The result is striking. Address heads are necessary, but the non-address support machinery is much more damaging when ablated as a group.

At 8k early position:

At 16k, the same qualitative pattern holds. Non-address core ablation is between -1.274 and -1.349 across the three answer positions, and query-tail ablation is between -0.946 and -1.043. Both are negative in 40/40 examples at every tested 16k setting. Address-head ablation is smaller but consistently negative, between -0.120 and -0.190 at 16k.

This means that support heads are not decorative. They are not merely heads that happen to be active near retrieval. Removing them reliably damages the model's correct-answer probability.

10.4 Matched inactive controls

The inactive controls are important because the non-address groups are larger than the address group. Without controls, one might argue that the non-address core is more damaging simply because it contains more heads.

The controls do not support that explanation. In the 8k early-position run, active non-address core ablation produces a mean delta of -1.269, while inactive controls are near zero or even slightly positive. Query-tail support produces -1.015, while its inactive control is only +0.024. This gap is too large to explain as generic late-layer ablation volume.

The same argument applies across positions and lengths. The active groups repeatedly damage answer probability, while matched inactive controls do not behave like the active circuit components.

10.5 Interpretation

The evidence so far supports a two-role view of the semantic retrieval circuit:

1. **Answer-address heads** directly attend to the answer-bearing context and are causally relevant.
2. **Support-state heads** are strongly necessary but do not directly attend to the answer span.

This is already a stronger explanation than the previous submission. The previous report established that retrieval-like heads exist and matter. The current result shows that the retrieval mechanism is internally differentiated: the model needs both address/content pathways and support-state machinery.

11 RESULTS III: L22H7 IS THE DOMINANT ANSWER-CONTENT HEAD

The previous section showed that address heads and support heads have different causal roles. The next question is whether answer content is distributed evenly across the address heads or concentrated in one head. Clean-to-corrupt patching and single-head decomposition show that answer content is strongly concentrated in L22H7.

11.1 Group patching separates content donors from support heads

In clean-to-corrupt patching, the corrupt prompt changes the answer value, making the clean answer unlikely. Patching asks whether copying selected clean activations into the corrupt run restores probability for the clean answer.

At 8k early position, the recovery gap is about 3.094 log-probability units: the clean answer has mean log probability -2.012 under the clean prompt, but only -5.106 under the corrupt prompt. This gives a meaningful space for recovery.

The answer-address group patches strongly:

$$\Delta_{\text{patch}}(\text{answer-address}) = +0.508.$$

Group	Heads	Mean ablation delta	Negative examples
Answer-address	3	-0.163	39/40
Non-address core	16	-1.269	40/40
Query-tail support	9	-1.015	40/40
First-token/sink	6	-0.376	40/40
Address inactive control	3	+0.011	13/40
Query-tail inactive control	9	+0.024	19/40

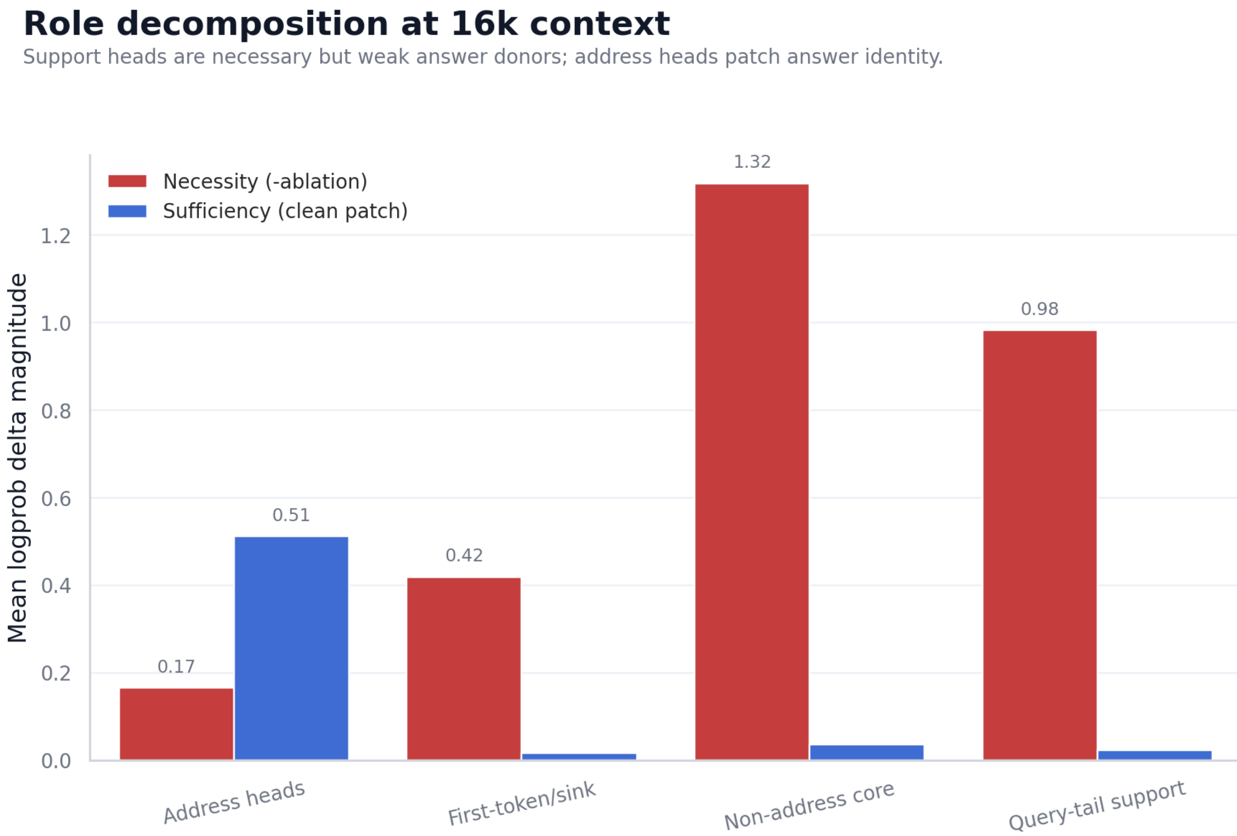

Figure 1: Role decomposition at 16k context. Support heads are strongly necessary under ablation but weak clean-patch answer donors; address heads are the main answer-content donors.

The non-address core, despite being much more damaging under ablation, barely patches:

$$\Delta_{\text{patch}}(\text{non-address core}) = +0.035.$$

Query-tail and first-token/sink groups are even smaller, around +0.013 and +0.011 in the same setting.

This is the key necessity/sufficiency split. Non-address support heads are necessary for the model's retrieval behavior, but their clean activations do not transplant the clean answer into the corrupt prompt. Address heads, by contrast, carry recoverable answer-specific signal.

At 16k, the same pattern holds. Answer-address patching remains large: +0.461, +0.538, and +0.535 across early, middle, and late positions. Non-address core patching remains tiny: +0.024, +0.041, and +0.043. At 16k, answer-address patching is roughly 12x to 19x larger than non-address core patching.

11.2 Single-head patching identifies L22H7

The answer-address group contains L22H7, L22H10, and L21H11. Single-head patching decomposes this group.

At 8k early position:

The full answer-address group patch is +0.508, so L22H7 alone accounts for about 90% of the group-level effect in this setting. Across all tested 8k and 16k settings, L22H7 accounts for roughly 88% to 94% of the answer-address group patch effect.

This is the strongest head-level result in the project. L22H7 is not merely one address head among several. It is the dominant clean answer-content donor under the patching operation.

Head	Patch delta	Recovery fraction	Positive examples
L22H7	+0.457	15.2%	38/40
L22H10	+0.072	2.4%	36/40
L21H11	+0.016	0.6%	25/40

11.3 L22H10 is a smaller companion

L22H10 also contributes positively, but its effect is much smaller. It patches between about +0.051 and +0.111 across the tested settings. Its relative role is most visible in some semantic variants, especially alias and relational prompts. This suggests that L22H10 may carry a secondary answer-related signal or participate in a companion address route.

The important point is that L22H10 supports the answer-content story without displacing L22H7. We therefore treat it as a companion head, not as an equal partner.

11.4 L21H11 is address-like but weak as a donor

L21H11 is a useful nuance. It often attends strongly to the answer span and is necessary under ablation, but it is weak as a standalone clean donor. Its patch effect stays near +0.016 to +0.032 across the tested settings.

This matters because it prevents an oversimplified interpretation: direct answer attention does not automatically imply strong transplantable answer content. L21H11 may help route, select, or prepare answer information without carrying the main clean-answer identity in the patched representation.

11.5 Activation-sensitive controls

The experiments also include L20H5 as an activation-sensitive control. It shows a noticeable clean-corrupt activation difference in some analyses, but it does not behave like a meaningful answer-content donor. Its patch effect is near zero or small compared with L22H7.

This control is valuable because it shows that activation sensitivity alone is not enough. A head can change when the prompt answer changes without being a major causal carrier of the answer. The final claim therefore rests on the convergence of attention, activation, ablation, and patching, not on activation difference alone.

11.6 Interpretation

The evidence supports the following decomposition:

- L22H7 is the dominant answer-content head.
- L22H10 is a smaller companion answer head.
- L21H11 is address-like and necessary but weak as a standalone answer donor.
- Support heads such as L20H7, L18H3, L17H10, L22H0, and L22H4 are necessary but not content donors under this patching operation.

This result is what turns the project from a retrieval-head atlas into a retrieval-circuit explanation. We can now say not only that some heads matter, but which head carries most of the transplantable answer identity.

12 RESULTS IV: ATTENTION, ACTIVATION, AND CAUSALITY CONVERGE ACROSS 8K-16K CONTEXTS

The strongest interpretability claims come from converging evidence. A head is more convincing as a mechanism when it has the right attention pattern, the right causal effect, the right activation geometry, and the same behavior across settings. L22H7 satisfies all four conditions.

12.1 L22H7 remains positive across all tested settings

L22H7 single-head patching is positive in every tested context length and answer position:

The confidence intervals remain above zero in all six settings. L22H7 is therefore not a one-position or one-length artifact.

The only notable wrinkle is the 8k middle-position dip, where L22H7 patching falls to +0.335. This is still strongly positive, but smaller than early and late positions. Rather than hiding this, we use it to show that the mechanism is robust but not perfectly invariant. Real mechanisms often have such texture.

12.2 L22H7 directly attends to the needle

L22H7 also has strong attention evidence. Its needle attention mass is high in every setting:

In all settings, the head places most of its query-step attention mass on the marked needle span, and substantial mass on the gold answer tokens themselves. This strongly supports the interpretation that L22H7 is an answer-address head.

12.3 L22H7 changes when the answer changes

Activation-difference analysis adds another layer of evidence. L22H7 has large clean-corrupt activation relative differences:

This means L22H7's query-step representation changes strongly when the answer identity changes. This is exactly what we would expect from a head that carries answer-specific information.

The contrast with support heads is important. L20H7, one of the strongest necessary support heads, has very small activation relative differences, around 0.031 to 0.040 across the 16k and 8k settings. It is necessary, but it is not changing much with answer identity. This supports the distinction between answer content and support state.

12.4 Support necessity remains stable

The support-head result also generalizes. Non-address core ablation is negative in 40/40 examples at every tested setting, with mean deltas around -1.20 to -1.36 at 8k and -1.27 to -1.35 at 16k. Query-tail ablation is also negative in 40/40 examples at every tested setting, with mean deltas around -0.95 to -1.04.

At the same time, non-address core patching remains tiny: roughly +0.024 to +0.043 at 16k, compared with answer-address patching of

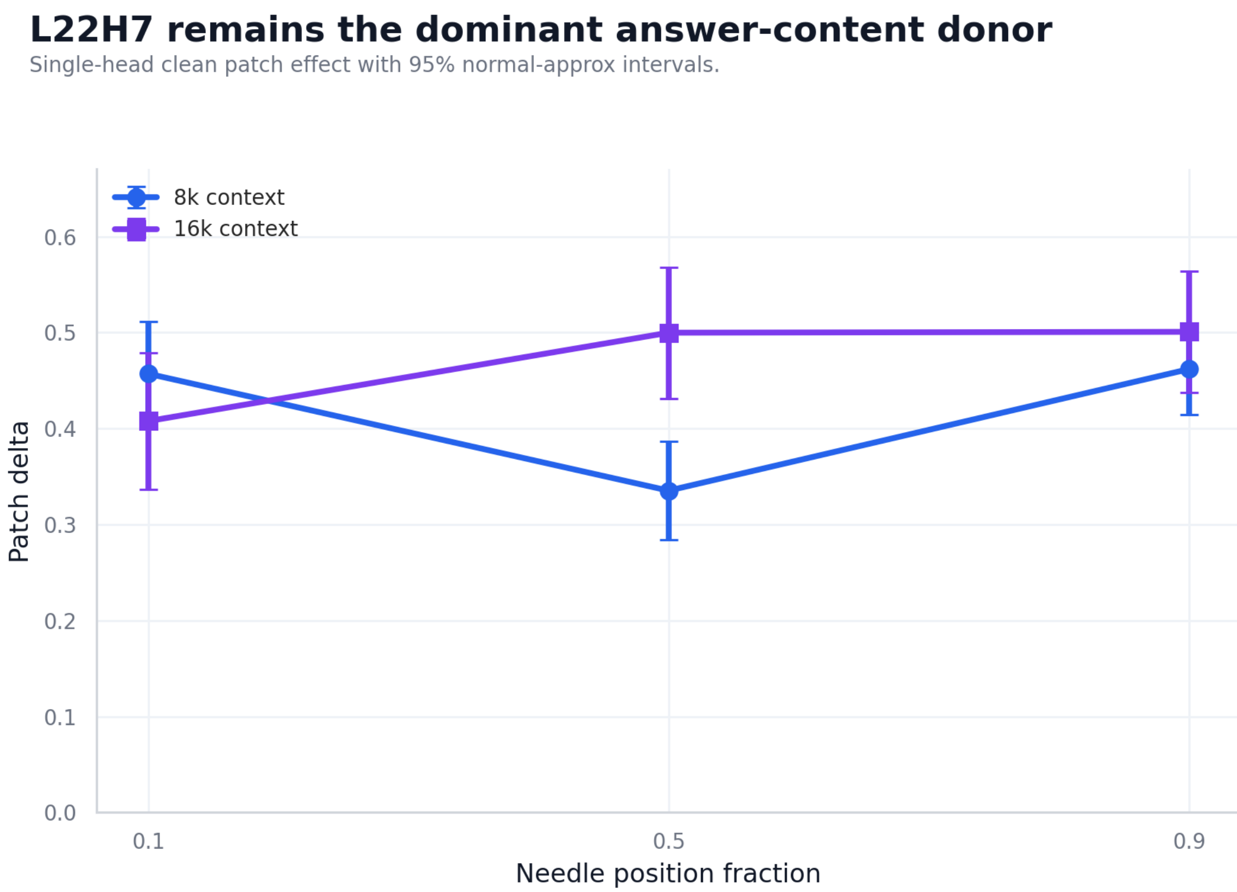

Figure 2: L22H7 remains the dominant single-head answer-content donor across context length and answer position. Error bars show 95% normal-approximation intervals.

Setting	L22H7 patch delta	95% CI
8k / 0.1	+0.457	[+0.403, +0.512]
8k / 0.5	+0.335	[+0.284, +0.387]
8k / 0.9	+0.462	[+0.414, +0.509]
16k / 0.1	+0.408	[+0.337, +0.479]
16k / 0.5	+0.500	[+0.431, +0.568]
16k / 0.9	+0.501	[+0.438, +0.564]

Setting	Gold attention	Needle attention
8k / 0.1	0.665	0.848
8k / 0.5	0.521	0.805
8k / 0.9	0.690	0.894
16k / 0.1	0.594	0.865
16k / 0.5	0.666	0.919
16k / 0.9	0.673	0.931

+0.461 to +0.538. This repeated contrast is central to the final narrative:

- support heads are necessary,
- support heads are not strong answer donors,
- address heads carry the transplantable answer signal.

12.5 Evidence matrix

The final evidence matrix is useful because it compresses the result into one view. Across all six settings, the same rows stay high or low in the expected places:

- answer-address patch is consistently positive,
- L22H7 patch is consistently positive,

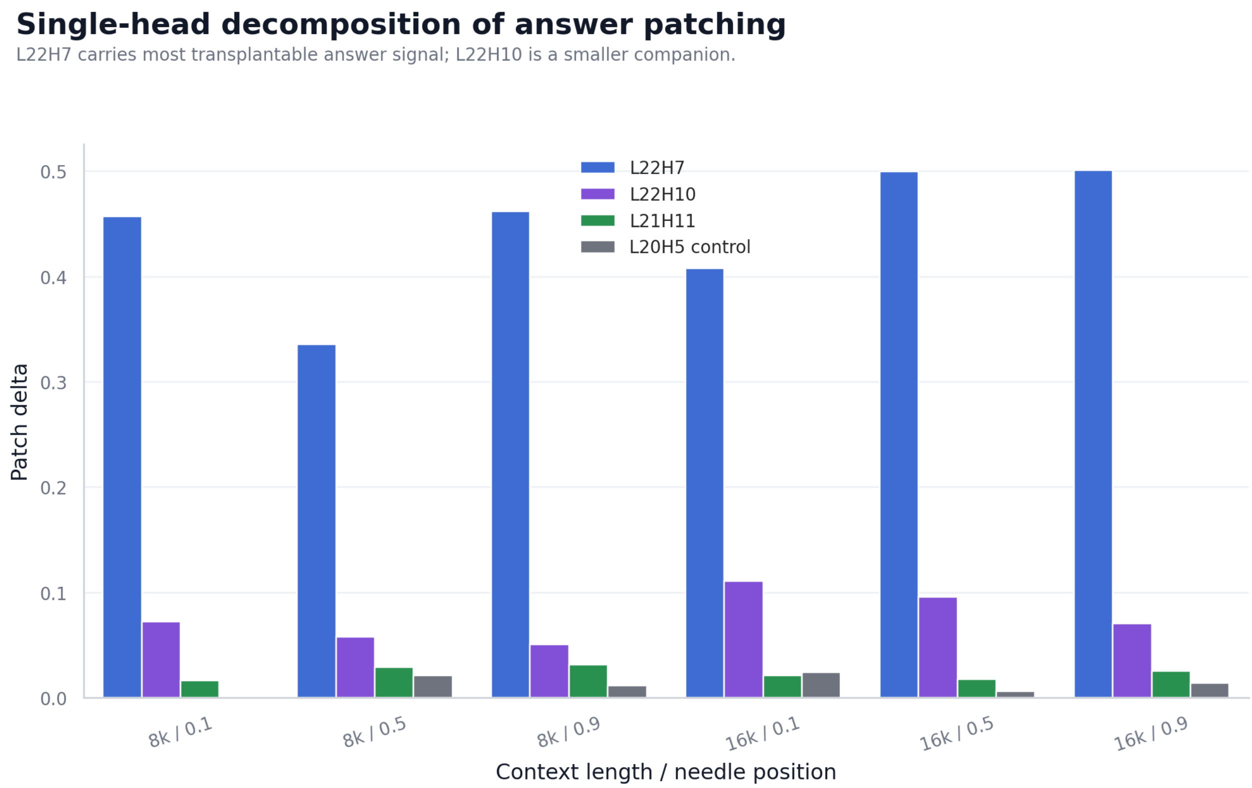

Figure 3: Single-head decomposition of answer patching. L22H7 carries most of the transplantable answer signal, while L22H10 is a smaller companion and controls remain small.

Setting	L22H7 activation relative difference
8k / 0.1	1.123
8k / 0.5	1.011
8k / 0.9	1.110
16k / 0.1	1.042
16k / 0.5	1.103
16k / 0.9	1.090

- support patch remains small,
- non-address and query-tail ablation necessity remains large,
- L22H7 needle attention remains high,
- L22H7 activation difference remains high.

This is the strongest "jury narrative" figure because it shows the volume of work without requiring the reader to inspect every CSV. It makes clear that the conclusion does not rest on one experiment.

Because the matrix combines different kinds of quantities, it should be read as an evidence map rather than as one shared physical scale. The important pattern is row-wise consistency across settings: content-patching rows stay positive, support-patching rows stay small, support-necessity rows stay large, and L22H7 attention/activation rows remain high.

12.6 Interpretation

The overall result is stable:

1. L22H7 attends to the answer-bearing span.
2. L22H7 changes strongly when the answer identity changes.
3. L22H7 causally restores clean-answer probability when patched.
4. L22H7 remains positive across 8k and 16k contexts.
5. Support heads remain necessary but not answer donors.

This convergence is the reason the final claim can be stronger than "we found a head with high attention." We found a head whose attention, activation geometry, and causal patch effect agree across a meaningful experimental grid.

13 DISCUSSION

13.1 From head discovery to circuit explanation

The previous submission discovered sparse retrieval-like heads and showed that selected heads had causal effects under ablation. The current work changes the level of explanation. Instead of only identifying heads, it decomposes the retrieval mechanism into roles.

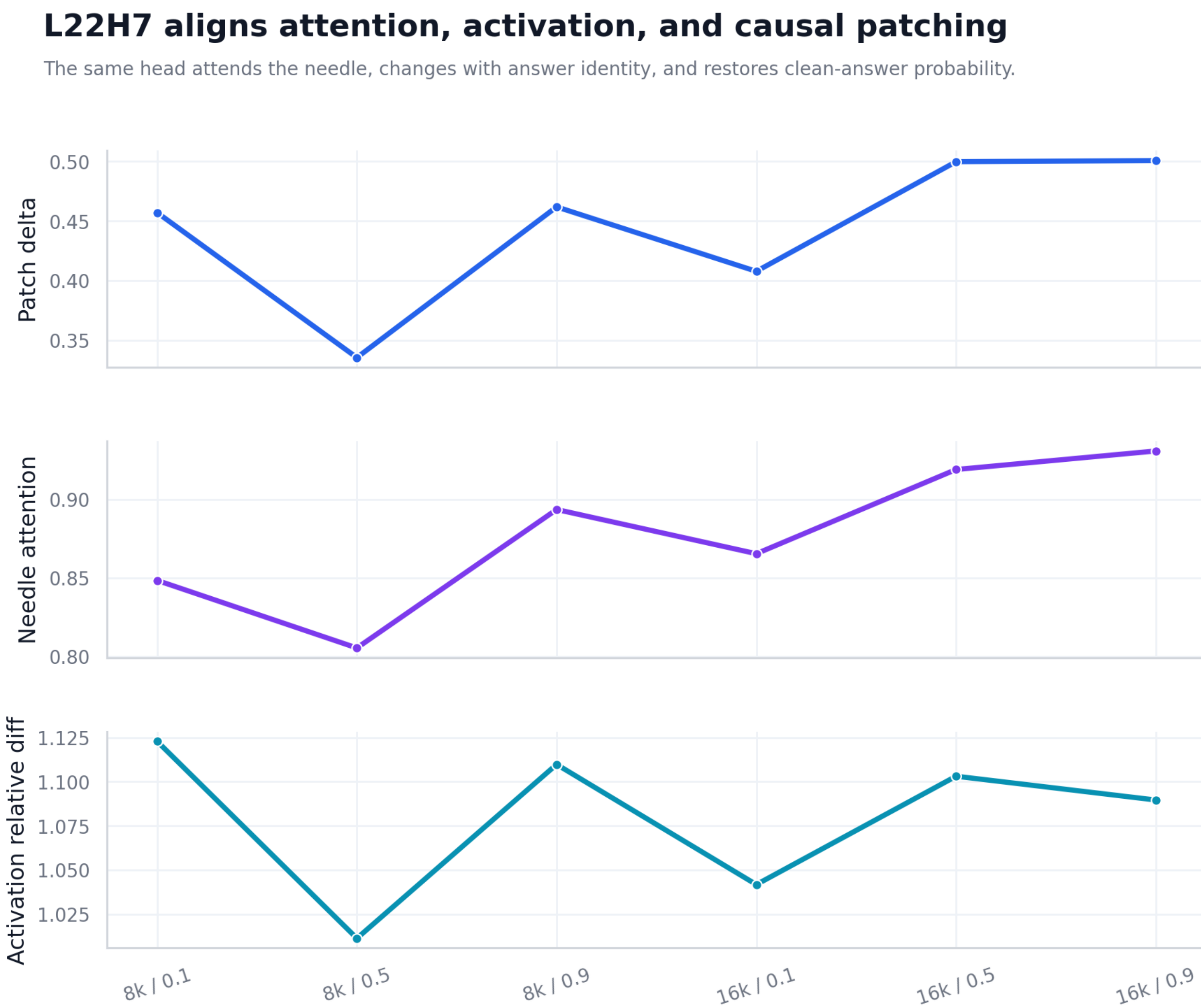

Figure 4: Converging evidence for L22H7. The same head attends to the needle, changes with answer identity, and restores clean-answer probability when patched.

The central interpretation is that semantic long-context retrieval in this model uses a role-decomposed circuit. One small address/content pathway carries answer identity, dominated by L22H7 and supported by L22H10. A broader set of support heads is required for retrieval performance but does not directly transplant answer identity under clean activation patching.

This is more informative than a simple ranking of heads. A head ranking says which components are important. A circuit explanation says how different components contribute.

13.2 Why support heads can be necessary without carrying the answer

The most important conceptual point is the separation between necessity and sufficiency. Non-address support heads are highly necessary: ablating them strongly reduces correct-answer log probability. But they are weak patch donors: copying their clean activations into a corrupt run barely restores the clean answer.

This makes sense if support heads maintain answer-independent retrieval state. They may shape the query representation, preserve useful residual directions, stabilize long-context computation, or prepare downstream components to use information from the address heads. If the clean and corrupt prompts differ only in answer identity, support-head activations may remain similar across the pair. Patching them then has little answer-specific information to transplant.

This explains the otherwise surprising contrast:

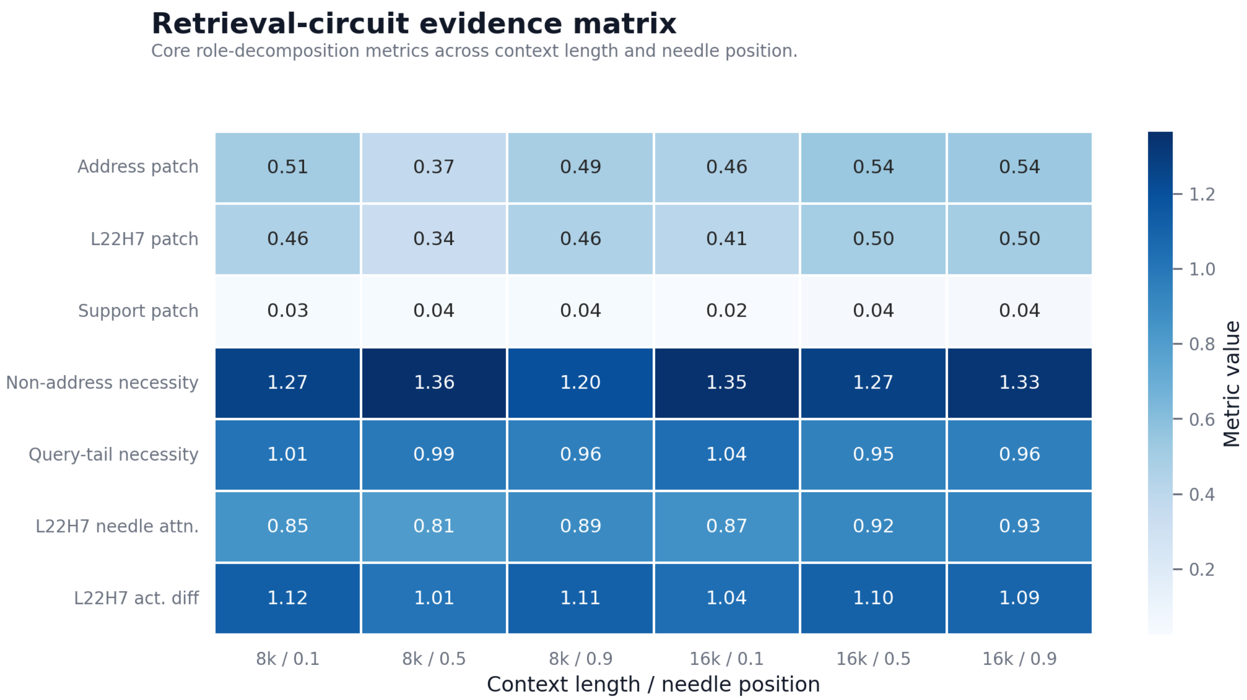

Figure 5: Retrieval-circuit evidence matrix across context length and needle position. Rows mix metric types, so the figure should be read row-wise as an evidence map rather than as one shared physical scale.

- non-address core ablation at 16k is around -1.27 to -1.35,
- non-address core patching at 16k is only around +0.024 to +0.043.

The support heads are doing work, but not the same work as L22H7.

13.3 Why attention alone is not enough

L21H11 shows why attention alone is insufficient. It often attends strongly to the answer span and is necessary under ablation, but its single-head patch effect is small. This means direct attention to the answer is not automatically equivalent to carrying the answer in a cleanly transplantable form.

Attention tells us where a head looks. Patching tests what its activation can do. Ablation tests whether the computation needs it. A convincing mechanism should align these signals, but they do not have to align perfectly for every head.

L22H7 is compelling because all three signals align. L21H11 is instructive because they do not fully align.

13.4 Why activation difference alone is not enough

Activation-difference analysis is also not sufficient by itself. L20H5 shows that a head can change between clean and corrupt prompts without becoming a major causal answer donor. This is why the report does not claim “large activation difference equals mechanism.” Instead, activation difference is used diagnostically, together with ablation and patching.

The strongest claim belongs to L22H7 because it satisfies all criteria:

- direct needle attention,
- large clean-corrupt activation shift,
- positive patch effect,
- stability across settings.

13.5 Relation to the previous patching result

The previous submission found that random patching could sometimes outperform patching the ablation-selected head set. At the time, this suggested that the patching operator was broad and that many sites could carry enough information to repair answer probability.

The current work clarifies that result. The earlier patching experiment used head sets selected for retrieval-event or ablation-style importance. But necessity-oriented selection and sufficiency-oriented selection need not identify the same heads. Once we decompose the circuit, the patching result becomes cleaner: L22H7, an answer-address/content head, carries most of the transplantable answer signal. Support heads can be necessary without patching strongly.

So the earlier “patching is broad” observation was not wrong. It was incomplete. The present work gives a more precise patching story by separating answer-content heads from support heads.

13.6 What the result says about semantic retrieval

Within this model and task family, semantic retrieval appears to involve a small content pathway plus broader support machinery. The model does not simply search the context with every head. It also does not appear to store the answer uniformly across all necessary heads. Instead, the answer identity is concentrated in a small number of heads, especially L22H7, while other heads make the retrieval computation usable.

This is a meaningful interpretability result because it connects several levels of evidence:

- semantic robustness across prompt variants,
- causal necessity under ablation,
- causal sufficiency under clean-to-corrupt patching,
- attention alignment with the source span,
- activation geometry under answer changes,
- position and context-length generalization.

We therefore do not claim that every model uses this exact head or this exact circuit. The defensible claim is model- and task-specific: Qwen2.5-1.5B-Instruct uses a stable role-decomposed circuit for this controlled semantic long-context retrieval family.

14 LIMITATIONS AND FUTURE WORK

14.1 Single-model scope

All experiments were run on Qwen2.5-1.5B-Instruct. This makes the study internally coherent but limits external validity. We do not claim that L22H7, or even the same layer/head structure, will appear in other models. A natural next step is to run the same pipeline on another Qwen size and at least one different model family.

The deeper question is whether the role decomposition generalizes, not whether the literal head index generalizes. It would still be a strong result if other models had different dominant content heads but similar separation between answer-content and support-state heads.

14.2 Synthetic task family

The semantic retrieval prompts are controlled and useful, but synthetic. They let us know the exact answer span, distractor span, prompt length, and clean/corrupt answer identity. This control is necessary for mechanistic experiments, but it does not cover all natural retrieval settings.

Future work should add a naturalistic retrieval task where the answer is embedded in more realistic documents. The strict non-truncation idea from the previous submission remains relevant: naturalistic long-context evaluation should avoid silently cutting the prompt, because truncation changes the task.

14.3 Patch-site limitation

The patching experiments intervene at the query-step input to the attention output projection. This is a useful and interpretable site, but it is not the only possible site. It does not separate query, key, value, attention logits, value vectors, or MLP contributions.

Future experiments could patch:

- attention logits,
- value vectors,
- Q/K/V projections separately,
- residual stream states before and after attention,
- multiple timesteps rather than only the query step.

These experiments would help answer whether L22H7's answer-content role is mainly in where it attends, what values it reads, or what it writes into the residual stream.

14.4 Dataset size and uncertainty

The main grid uses 40 examples per setting. The effects are strong and consistent enough for the present claim, especially because many signs hold in 38/40, 39/40, or 40/40 examples. Still, a final paper version would benefit from larger confirmation runs, especially for the most important settings.

A high-N confirmation run could use the same pipeline with more examples per variant at 16k. This would tighten confidence intervals and make the statistical presentation stronger.

14.5 Possible failure modes

The current prompt family tests literal, alias, paraphrase, relational, and distractor-heavy retrieval. It would be useful to find where the circuit stops working or changes form. Possible stress tests include:

- multiple answer-bearing spans with conflicting relations,
- longer and more natural distractor passages,
- answers that are not six-digit strings,
- multi-hop queries requiring two or more retrieved facts,
- adversarial paraphrases where the query is semantically close but not identical.

Such failure modes could turn into a strong extension: not only "this is the circuit when retrieval succeeds," but "this is how the circuit changes or fails when retrieval becomes harder."

15 CONCLUSION

This report extends the Retrieval Head Atlas from literal long-context copying to semantic long-context retrieval. The previous submission showed that retrieval-like behavior is sparse and causally localized. The current work shows that the mechanism is also role-decomposed.

Across these controlled semantic prompt variants, answer positions, and 8k-16k context lengths, Qwen2.5-1.5B-Instruct uses a stable retrieval circuit with separable roles. L22H7 is the dominant answer-content head: it attends to the answer-bearing span, changes strongly when the answer changes, and restores clean-answer probability when patched into a corrupt run. L22H10 contributes as a smaller companion. L21H11 is address-like and necessary but weak as a standalone answer donor. A broader set of non-address support heads is strongly necessary under ablation but does not itself transplant answer identity under clean activation patching.

The main takeaway is simple:

> Semantic long-context retrieval in Qwen2.5-1.5B-Instruct is not just a diffuse model capability and not just a list of important heads. It is supported by a role-decomposed circuit: a small address/content

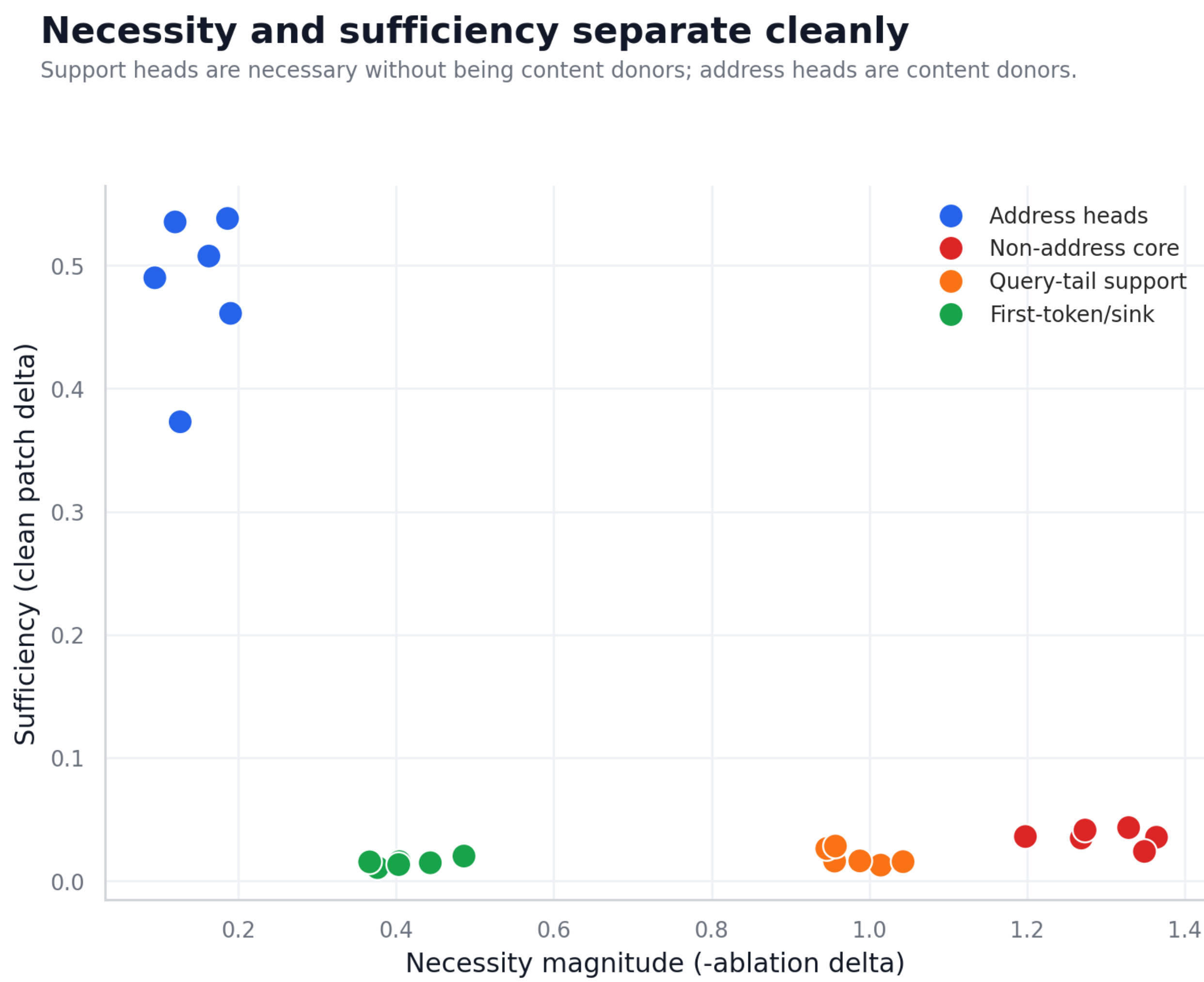

Figure 6: Necessity and sufficiency separate cleanly. Support heads are necessary without being strong answer donors; address heads are the meaningful content donors.

pathway carries answer identity, while broader support heads make the retrieval computation work.

This gives the project a stronger final narrative than the previous submission. We began by mapping retrieval heads. We now have evidence for a retrieval circuit.

The appendix keeps the main report readable while preserving the implementation map needed to reproduce and audit the work.

A APPENDIX A: GLOSSARY

- **Attention head.** One parallel attention component inside a transformer layer. Each head computes its own attention pattern and writes a head-specific vector back into the residual stream.
- **KV cache.** Cached key and value tensors from earlier prompt tokens, used so the model can score the final query and answer tokens without recomputing the whole prefix.
- **Retrieval head.** An attention head that attends to answer-relevant context and has causal evidence for affecting retrieval behavior.
- **Answer-address head.** A retrieval head whose query-step attention directly targets the answer-bearing needle span.
- **Support head.** A head that is necessary under ablation but does not directly carry the answer identity under the clean-to-corrupt patching operation.
- **Ablation.** An intervention that removes a selected component, here by zeroing selected head slices at the query-step attention output site.
- **Activation patching.** An intervention that copies internal activations from a clean run into a corrupted run to test whether that component carries information sufficient to repair the output distribution.
- **Clean/corrupt prompt pair.** A paired prompt construction where the clean prompt contains the target answer

and the corrupt prompt changes the answer value while preserving the rest of the retrieval structure.
- **Teacher-forced log probability.** The mean log probability assigned to the gold answer tokens when the answer tokens are supplied one by one.
- **Recovery fraction.** The fraction of the clean-corrupt log-probability gap restored by a patching intervention.
- **Attention mass.** The total attention probability assigned by a head to a specified token span, such as the gold answer span or the full needle span.
- **Activation relative difference.** A normalized measure of how much a head activation changes between clean and corrupt prompts.
- **Layer-matched control.** A random control group that samples the same number of heads from the same layers as the tested group.
- **Inactive control.** A control head or head group selected because prior experiments showed weak activity under the relevant retrieval metrics.

B APPENDIX B: REPRODUCIBILITY NOTES

The main experiment scripts are:

- `scripts/run_semantic_ablation_probe.py`
- `scripts/run_semantic_single_head_sweep.py`
- `scripts/run_semantic_attention_trace.py`
- `scripts/run_semantic_group_ablation.py`
- `scripts/run_semantic_component_patching.py`
- `scripts/run_semantic_patch_interaction.py`
- `scripts/run_semantic_activation_delta.py`
- `scripts/run_position_generalization.sh`
- `scripts/build_report_assets.py`

The main report assets are generated from the committed experiment outputs using `scripts/build_report_assets.py`. The LaTeX report is generated locally from `docs/submission_report_draft.md`, `paper/main.tex`, and `paper/references.bib` using `make report-pdf` and `make report-pages`.

The expected project workflow is: code is edited locally, GPU experiments are run on the SageMaker notebook instance, artifacts are pulled back into the repository, and report assets are regenerated locally from those committed artifacts.

C APPENDIX C: REPORT ASSET MAP

The report uses six generated body figures:

- `fig_01_role_decomposition_16k.png`: role decomposition between necessity and sufficiency.
- `fig_02_l22h7_generalization.png`: L22H7 patching across length and position.
- `fig_03_single_head_decomposition.png`: L22H7 compared with other address heads and controls.
- `fig_04_l22h7_attention_activation_alignment.png`: attention, activation, and patching alignment for L22H7.
- `fig_06_evidence_matrix.png`: compact evidence matrix across all tested settings.
- `fig_05_necessity_vs_sufficiency.png`: conceptual separation between support heads and content donors.

The generated tables are stored with the report assets as CSV, Markdown, and LaTeX files. They are used to audit the numerical claims in the main report and can be included in a longer appendix if required by the submission format.

D APPENDIX D: REFERENCE MAP

The report cites the following core reference areas:

- Transformer architecture: Vaswani et al., “Attention Is All You Need”
- Qwen2.5 model: Qwen2.5 technical report
- Retrieval heads / long-context factuality: “Retrieval Head Mechanistically Explains Long-Context Factuality”
- Long-context position effects: “Lost in the Middle”
- Transformer circuits and induction heads: Anthropic transformer-circuits work and induction-head work
- Activation patching / causal tracing: ROME and related causal tracing work
- Tooling: Hugging Face Transformers [7]

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[7] Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Remi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. 2020. Transformers: State-of-the-Art Natural Language Processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. 38–45. https://doi.org/10.18653/v1/2020.emnlp-demos.6

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